

Shreyansh Kabra

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SUMMARY

Software Engineer and USC MSCS candidate with 2+ years of production experience across full-stack development, cloud infrastructure, and distributed systems, supported by cloud-scale expertise gained at AWS.

PROFESSIONAL EXPERIENCE

Software Development Engineer Intern

May 2026 - Aug 2026 (Expected)

Amazon Web Services (Amazon Connect)

Seattle, WA

- Designed and shipped a distributed Redis L2 cache (ElastiCache Serverless) beneath a per-host in-memory L1 (Caffeine) cache for Amazon Connect's real-time agent-routing metrics, cutting redundant downstream fetches by 90% (118K to 12K calls per cycle) across a 591-host fleet.
- Built a fault-tolerant read-through client (Lettuce) with a 100ms command timeout and a write-guard against cache poisoning, sustaining single-digit-millisecond p99 reads with clean fallback to source systems on any cache miss or failure.
- Gated the launch behind a DynamoDB-backed dynamic feature flag (instant kill switch, sub-60s rollback) emitting CloudWatch hit/miss/stale metrics, and validated it with unit and integration tests for a zero-downtime rollout.

Cloud Developer 1

Aug 2022 - Dec 2024

Hewlett Packard Enterprise

Bangalore, India

- Built 100+ RESTful APIs using Python, Django, and DRF for the GreenLake edge-to-cloud platform, driving high-volume synchronization between PostgreSQL and ServiceNow instances.
- Architected an asynchronous email notification system (API + UI), decoupling message processing from the request path and lifting operational efficiency by 80%.
- Reworked cross-module integrations for the Service Insights Portal, cutting submission errors by 50% through stronger validation logic and mentoring 3 junior developers on API design.

R&D Intern, NaaS Team

Jan 2022 - Aug 2022

Hewlett Packard Enterprise

Bangalore, India

- Developed an asynchronous Python package to automate network device calibration, removing over 90% of manual work and improving reliability at scale.
- Streamlined troubleshooting report generation and execution, trimming task completion time by 50% and manual effort by 70%.

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, C++, SQL, NoSQL, HTML5, CSS3, Django, React.js, Node.js

Cloud, Tools & AI: AWS (ElastiCache, CloudWatch), Google Cloud Platform (GCP), Git, Postman, AWS Kiro, Claude Code, Ollama

Databases & Concepts: PostgreSQL, MongoDB, MySQL, Redis, REST APIs, Microservices, Distributed Systems, Data Structures & Algorithms, System Design, RAG

EDUCATION

Master of Science, Computer Science

Jan 2025 - Dec 2026 (Expected)

University of Southern California

Los Angeles, CA

Bachelor of Technology, Computer Science and Engineering

Aug 2018 - May 2022

JECRC University

Jaipur, India

ACADEMIC PROJECTS

Hallucination-Resistant Multi-Hop QA Pipeline |Python, FAISS, BM25, Ollama, HuggingFace

Jan 2026 - May 2026

- Constructed a seven-stage RAG pipeline for multi-hop question answering on HotpotQA (7,400+ examples), fusing BM25 and FAISS retrieval via Reciprocal Rank Fusion with iterative two-hop entity extraction.
- Added cross-encoder reranking and a post-generation verification layer (NLI, lexical overlap), reaching 68% Exact Match and 80% F1 while curbing hallucination across 4,800+ lines of Python.

Mental Health Signal Detector |NLP, RAG, RoBERTa, FAISS, Ollama

Aug 2025 - Dec 2025

- Fine-tuned RoBERTa-Large with LoRA to classify mental-health intent and concern levels, scoring 0.81 F1 within a hierarchical NLP framework optimized for memory efficiency.
- Paired FAISS vector search with Flan-T5 in a retrieval-augmented system to generate grounded, safety-compliant support suggestions from 6,000+ curated samples.

Artist Search Platform |React.js, Node.js, MongoDB

Mar 2025 - May 2025

- Crafted a full-stack app (React.js web + native Android in Kotlin) on a Node.js/Express backend, serving Artsy data on 100,000+ artists and artworks.
- Secured access with JWT authentication over structured MongoDB collections, achieving 100% Lighthouse scores on mobile and desktop.